Solutions for Real Analysis Homework 2

Problem 1. Let S be the semi-ring of all intervals in $[0,1]$, and $g \in C[0,1]$ is a nondecreasing function. Prove that $\lambda(I) := g(b) - g(a)$ is a sigma-additive measure on S.

Solution. Let $I \in S$ and $\{I_k\}_{k=1}^{\infty} \subset S$ be a countable disjoint collection such that $I = \bigcup_{k=1}^{\infty} I_k$. We need to show $\lambda(I) = \sum_{k=1}^{\infty} \lambda(I_k)$.

1. **Finite additivity:** For any finite N, $\bigcup_{k=1}^{N} I_k \subset I$. Since g is nondecreasing, $\sum_{k=1}^{N} \lambda(I_k) \leq \lambda(I)$. Taking $N \to \infty$, we get $\sum_{k=1}^{\infty} \lambda(I_k) \leq \lambda(I)$.

2. **Countable subadditivity (σ-additivity):** We need to prove $\lambda(I) \leq \sum_{k=1}^{\infty} \lambda(I_k)$. Let $\epsilon > 0$. Since g is continuous on $[0,1]$, it is uniformly continuous. For $I = [a,b)$ (w.l.o.g.), pick $\delta > 0$ such that $g(b) - g(b-\delta) < \epsilon$. Let $I' = [a, b-\delta]$, which is a compact set. For each $I_k = [a_k, b_k)$, pick $\eta_k > 0$ such that $g(b_k) - g(a_k - \eta_k) < \lambda(I_k) + \epsilon/2^k$. Let $J_k = (a_k - \eta_k, b_k)$. The family $\{J_k\}$ forms an open cover of the compact set I'. By the Heine-Borel theorem, there exists a finite subcover $\{J_{k_1}, \ldots, J_{k_m}\}$. Using the elementary subadditivity on finite covers:
$$\lambda(I) - \epsilon < \lambda(I') \leq \sum_{j=1}^{m} \lambda(J_{k_j}) < \sum_{k=1}^{\infty} (\lambda(I_k) + \epsilon/2^k) = \sum_{k=1}^{\infty} \lambda(I_k) + \epsilon.$$
Since ϵ is arbitrary, $\lambda(I) \leq \sum_{k=1}^{\infty} \lambda(I_k)$.

□

Remark. A common mistake is to skip the Heine-Borel compactness argument and simply claim "sum of lengths equals total length". On a semi-ring, σ-additivity is non-trivial and requires continuity/compactness.

Problem 2. Let $F \subset \mathbb{R}^n$ be a closed set of positive measure. Without using the Continuum Hypothesis, show that F is continual.

Solution. Since $\mu(F) > 0$, F is non-empty. Since $F \subset \mathbb{R}^n$, $|F| \leq \mathfrak{c} = 2^{\aleph_0}$. We prove $|F| \geq \mathfrak{c}$. We construct an injection from the space of infinite binary sequences $\{0,1\}^{\mathbb{N}}$ into F. Since $\mu(F) > 0$, there exists a closed ball B such that $\mu(F \cap B) > 0$. **Construction:**

1. Let $K_\emptyset = F \cap B$.

2. We claim that any closed subset $K \subset F$ with $\mu(K) > 0$ contains two disjoint closed subsets K_0, K_1 both with positive measure. *Proof of claim:* Split the bounding box of K by a hyperplane. If the intersection with the hyperplane has positive measure, the measure is not σ-finite on hyperplanes (false in $\mathbb{R}^n$), so boundaries have measure 0. Thus the mass is split between two halves. At least one has positive measure. Actually, since $\mu(K) > 0$, we can find two disjoint balls B_1, B_2 intersecting K with positive measure (otherwise K would be concentrated on a point, i.e., measure 0).

3. Apply this inductively. For each finite sequence $s \in \{0,1\}^k$, construct disjoint compact sets $K_{s0}, K_{s1} \subset K_s$ with $\operatorname{diam}(K_{si}) \leq 2^{-k}$.

For any $\sigma \in \{0,1\}^{\mathbb{N}}$, the set $\bigcap_{n=1}^{\infty} K_{\sigma|_n}$ is a nested intersection of non-empty compact sets with vanishing diameter. By Cantor's Intersection Theorem, it contains exactly one point x_σ. Since sets at each level are disjoint, $\sigma \neq \sigma' \implies x_\sigma \neq x_{\sigma'}$. Thus $|F| \geq |\{0,1\}^{\mathbb{N}}| = \mathfrak{c}$. $\square$

Remark. Simply "picking points" or using $A_n = \{y : \|y - x\| < d/n\}$ is insufficient because $F \setminus A_n$ might not maintain the positive measure property easily without a geometric argument. The binary tree construction is standard.

Problem 3. Let $(\Omega, \mathcal{A}, \mu)$ be a space with a σ-finite measure, $\mu(\Omega) = +\infty$. Prove that Egorov Theorem necessarily fails on this space.

Solution. Since the space is σ-finite with infinite measure, there exists a countable partition $\Omega = \bigcup_{n=1}^{\infty} E_n$ with disjoint measurable sets satisfying $0 < \mu(E_n) < \infty$ (or at least bounded away from 0 is not required, but sum must be ∞). Define $f_n = \chi_{\cup_{k=n}^{\infty} E_k}$.

- **Pointwise convergence:** For any $x \in \Omega$, x belongs to some specific E_m. For all $n > m$, $x \notin \bigcup_{k=n}^{\infty} E_k$, so $f_n(x) = 0$. Thus $f_n \to 0$ pointwise everywhere.
- **Uniform convergence:** Egorov's theorem asserts that for any $\epsilon > 0$, there exists A with $\mu(\Omega \setminus A) < \epsilon$ such that $f_n \to 0$ uniformly on A. Suppose such an A exists. Since $\mu(\Omega \setminus A) < \infty$ and $\mu(\Omega) = \sum \mu(E_k) = \infty$, A must have infinite measure. Specifically, A must intersect $\bigcup_{k=n}^{\infty} E_k$ for every n (otherwise A would be contained in a finite union of finite measure sets). Thus, $\sup_{x \in A} |f_n(x) - 0| = 1$ for all n. Convergence is not uniform.

$\square$

Problem 4. Based on properties of integrals on spaces with a finite measure, give a proof of the linearity of the Lebesgue integral in the setting of σ-finite complete measures.

Solution. Let $\Omega = \bigcup X_n$ with $X_n \nearrow \Omega$ and $\mu(X_n) < \infty$. Let $f, g \in L^1(\Omega)$ (integrable functions).

1. **Non-negative case:** If $f, g \geq 0$, then by Monotone Convergence Theorem (MCT):

$$\int_\Omega (f+g) = \lim_{n\to\infty} \int_{X_n} (f+g) = \lim_{n\to\infty} \left(\int_{X_n} f + \int_{X_n} g \right) = \int_\Omega f + \int_\Omega g.$$

   (Using linearity on finite measure space X_n).

2. **General case:** Note that $f + g = (f+g)^+ - (f+g)^-$. Also $f + g = (f^+ - f^-) + (g^+ - g^-)$. Rearranging terms to avoid subtraction of infinities (though here integrals are finite):

$$(f+g)^+ + f^- + g^- = (f+g)^- + f^+ + g^+.$$

   All functions are non-negative. Apply linearity for non-negative functions proved in step 1:

$$\int (f+g)^+ + \int f^- + \int g^- = \int (f+g)^- + \int f^+ + \int g^+.$$

   Since $f, g \in L^1$, all integrals are finite numbers. We can rearrange the equation:

$$\int (f+g)^+ - \int (f+g)^- = \left(\int f^+ - \int f^- \right) + \left(\int g^+ - \int g^- \right).$$

   Thus $\int (f+g) = \int f + \int g$.

$\square$

Remark. Writing $\int(f+g) = \int(f+g)^+ - \int(f+g)^- = \int f^+ + \int g^+ - \dots$ directly is risky without proving the rearrangement is valid. The safest way is to equate sums of positive parts first.

Problem 5. Construct an example of a sequence of Lebesgue integrable functions on $\mathbb{R}$ converging to 0 almost everywhere, but with integrals over $\mathbb{R}$ converging to ∞.

Solution. Define $f_n(x) = n^2\chi_{(0,1/n)}(x)$.

- **Pointwise:** For any $x \leq 0$, $f_n(x) = 0$. For $x > 0$, if $n > 1/x$, then $x \notin (0, 1/n)$, so $f_n(x) = 0$. Thus $f_n(x) \to 0$ for all x.
- **Integral:** $\int_{\mathbb{R}} f_n d\lambda = n^2 \cdot \frac{1}{n} = n$.
- **Limit:** $\lim_{n\to\infty} \int f_n = \lim n = \infty$.

$\square$

Problem 6. Let $(\Omega, \mathcal{A}, \mu)$ be a space with measure, and f an integrable on Ω function such that its integral over any measurable $A \subset \Omega$ vanishes. Prove that f is a.e. vanishing.

Solution. Let $E = \{x \in \Omega : f(x) > 0\}$. We assume $\mu(E) > 0$ and find a contradiction. Define $E_n = \{x \in \Omega : f(x) > 1/n\}$. Note that $E_n \nearrow E$. By continuity of measure, $\lim_{n\to\infty} \mu(E_n) = \mu(E)$. If $\mu(E) > 0$, there exists some N such that $\mu(E_N) > 0$. Choose $A = E_N$. This is a measurable set.

$$\int_{E_N} f d\mu \geq \int_{E_N} \frac{1}{N} d\mu = \frac{1}{N}\mu(E_N) > 0.$$

This contradicts the hypothesis that $\int_A f = 0$ for all A. Thus $\mu(\{f > 0\}) = 0$. Similarly, applying to $-f$, $\mu(\{f < 0\}) = 0$. Therefore, $f = 0$ a.e. $\square$

Problem 7. Let $\{q_n\}$ be a sequence of all rational numbers in $[0, 1]$, and $\sigma > 0$. Using the Lebesgue integration theory, prove that the series converges a.e.

Solution. Let $g(x) = \sum_{n=1}^{\infty} \frac{1}{n^{1+\sigma}\sqrt{|x-q_n|}}$. Since terms are non-negative, by Beppo Levi's Theorem (Monotone Convergence for series):

$$\int_0^1 g(x)dx = \sum_{n=1}^{\infty} \frac{1}{n^{1+\sigma}} \int_0^1 \frac{dx}{\sqrt{|x-q_n|}}.$$

Compute the inner integral:

$$\int_0^1 |x-q_n|^{-1/2}dx = \int_0^{q_n} (q_n - x)^{-1/2}dx + \int_{q_n}^1 (x-q_n)^{-1/2}dx = 2\sqrt{q_n} + 2\sqrt{1-q_n}.$$

Since $q_n \in [0, 1]$, $2\sqrt{q_n} + 2\sqrt{1-q_n} \leq 4$. Thus,

$$\int_0^1 g(x)dx \leq \sum_{n=1}^{\infty} \frac{4}{n^{1+\sigma}}.$$

Since $\sigma > 0$, the p-series converges, so $\int_0^1 g(x)dx < \infty$. A function with a finite integral must be finite almost everywhere. Thus the series converges a.e. $\square$

Problem 8. Investigate the Lebesgue integrability of $f(x) = x^\alpha \sin(x^\beta)$ on $[0,1]$ vs improper Riemann integral.

Solution. We analyze behavior as $x \to 0^+$.

1. **Case $\beta > 0$:** $\sin(x^\beta) \sim x^\beta$. $f(x) \sim x^{\alpha+\beta}$. Both integrals converge iff $\alpha + \beta > -1$.
2. **Case $\beta = 0$:** $f(x) = x^\alpha \sin(1)$. Both converge iff $\alpha > -1$.
3. **Case $\beta < 0$:** Let $\gamma = -\beta > 0$. $f(x) = x^\alpha \sin(x^{-\gamma})$. Change variables $t = x^{-\gamma} \implies x = t^{-1/\gamma}$. Integral becomes $\sim \int^\infty t^{-\frac{\alpha+1}{\gamma}-1} \sin(t)dt$.
   - **Lebesgue (Absolute Convergence):** Requires $\int^\infty |t^{-p} \sin t| < \infty$, which holds iff $p > 1$. $-\frac{\alpha+1}{\gamma} - 1 < -1 \implies \alpha > -1$.
   - **Riemann (Conditional Convergence):** Dirichlet test applies if $t^{-p} \searrow 0$. Requires exponent < 0. $-\frac{\alpha+1}{\gamma} - 1 < 0 \implies \alpha > -1 - \gamma = -1 + \beta$.

Conclusion for $\beta < 0$:

- $\alpha > -1$: Both converge.
- $-1 + \beta < \alpha \leq -1$: Improper Riemann converges, Lebesgue diverges.
- $\alpha \leq -1 + \beta$: Both diverge.

□

Remark. The case $\beta < 0$ with $\alpha \leq -1$ is crucial. This is where Lebesgue integration fails (infinite absolute area) but Riemann succeeds (cancellation of oscillating areas).

Problem 9. Prove the analogue of the Riemann conditional convergence theorem for integrals.

Solution. Given $\int_X |f| d\mu = \infty$ but $\lim_N \int_{[1,N]} f$ is finite. This implies the positive and negative parts diverge: $\int f^+ = \infty$ and $\int f^- = \infty$. Since the measure is non-atomic (Lebesgue measure on $[0,\infty)$), for any set E with infinite integral and any $M > 0$, we can find a subset $E' \subset E$ with $\int_{E'} f = M$ (Darboux property of the integral). Algorithm to construct A_j:

1. Let $P = \{f \geq 0\}$ and $M = \{f < 0\}$. Both have infinite integrals.
2. Pick disjoint bounded subsets $P_1, P_2, \ldots$ from P and $M_1, M_2, \ldots$ from M to mimic the terms of a conditionally convergent series.
3. Start with $A = \emptyset$. Add mass from P until $\int_A f > a$. Then add mass from M until $\int_A f < a$.
4. Crucially, to ensure $X = \bigcup A_j$, at step k, ensure that the interval $[0,k]$ is included in the current set A_k. If adding $[0,k]$ disturbs the sum, use the "infinite reservoir" of remaining P or M to correct the sum back towards a.

Since $\lim_{N\to\infty} \int_{[N,\infty)} f$ isn't necessarily 0 (it doesn't exist absolutely), we rely on the fact that f is locally integrable. We can finely control the integral value by adding small chunks of the domain. □

Problem 10. Show that $\int_\Omega f d\mu = \sup\left\{\int_\Omega g d\mu,\ 0 \leq g \leq f,\ g \text{ finite \& measurable}\right\}$.

Solution. Let $I = \int_\Omega f d\mu$ and S be the supremum.

- $S \leq I$: For any $0 \leq g \leq f$, by monotonicity, $\int g \leq \int f$. Thus $\sup \int g \leq \int f$.
- $I \leq S$: By definition of the Lebesgue integral for $f \geq 0$, there exists a sequence of simple functions s_n such that $0 \leq s_n \nearrow f$. By Monotone Convergence Theorem, $\lim \int s_n = \int f$. Simple functions are finite-valued and measurable. Thus s_n are candidates for g. Since $S \geq \int s_n$ for all n, we have $S \geq \lim \int s_n = I$.

Therefore, $I = S$. $\square$